



Society of Petroleum Engineers

SPE-229433-MS

Characterizing and Predicting High-Permeability Flow Zones in the Shuaiba Formation of a Giant Carbonate Onshore Oilfield Through Stratigraphic, Diagenetic, and Dynamic Integration

K. M. Torres, Abu Dhabi National Oil Company, Abu Dhabi, Abu Dhabi, UAE / Departament de Mineralogia, Petrologia i Geologia Aplicada, Facultat de Ciències de la Terra, Universitat de Barcelona, c/ Martí i Franquès s/n, Spain; H. Elsahn, R. Gutierrez, N. Al Madani, O. Ardelean, and H. Al Naqbi, Abu Dhabi National Oil Company, Abu Dhabi, Abu Dhabi, UAE

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/229433-MS](https://doi.org/10.2118/229433-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

This study examines the high permeability flow zones (HPFZs) on reservoir performance in the Shuaiba Formation of a giant carbonate onshore oilfield located in Abu Dhabi, UAE. The objective is to characterize and predict HPFZs by enhancing the stratigraphic framework and permeability architecture through the integrated sedimentological, petrophysical, and dynamic data. The study focuses on HPFZs resulting from preserved primary porosity, diagenetic enhancement, and structurally induced fracturing associated within fault systems to assess the impact of these features on waterflood performance and support the optimization of hydrocarbon recovery.

A multi-disciplinary, data driven workflow was implemented combining core description, borehole image log interpretations, well test analyses, and production log data across more than 500 wells. High-resolution stratigraphic correlations were used to delineate depositional architectures and facies stacking patterns, which were then calibrated with pressure and saturation trends. HPFZs were recognized through contrasts between core permeability and dynamic responses from PLT and well test data. Structural features such as fault-related fracture corridors were assessed using borehole image logs and seismic attributes to enable spatial prediction of fluid conduits and barriers. This integration enabled a clearer understanding of the spatial extent and interaction of HPFZs.

Analysis of the Shuaiba Formation reveals that a small number of laterally continuous, high-permeability rudist-rich facies, account for a high permeability HPFZs and subsequently disproportionate share of fluid production and injection response. These HPFZs are frequently associated with shallower depositional periods and regressive parasequences and exhibit multi-Darcy permeabilities compared to the background matrix (<10 mD). Diagenetic overprinting, including leaching, was observed to further enhance permeability in localized zones. Structural mapping indicates that subtle fault damage zones occasionally coincide with HPFZ pathways, enhancing vertical connectivity and exacerbating early water breakthrough in crestal producers. A significant permeability contrast between HPFZs and surrounding

matrix rock contributes to poor flood conformance, evidenced by bypassed pay and irregular saturation trends. Integration of dynamic surveillance with stratigraphic modeling enabled the refinement of the static model to account for HPFZ influenced flow behavior. These results support the need to condition simulation models with deterministic HPFZ distributions, particularly in the southern area, near the Graben Region in the central area and crestal sectors of the field where water encroachment is most severe. These findings underscore the necessity of incorporating HPFZs deterministically into simulation models to minimize conformance issues, reduce bypassed pay, and improve recovery factors

This work delivers the first integrated stratigraphic-dynamic-structural HPFZ mapping of the Shuaiba Formation in the giant oilfield. By integrating depositional architecture, diagenetic trends, and dynamic flow diagnostics, it introduces a predictive framework for HPFZ identification and incorporation into reservoir models, offering a scalable methodology for optimizing recovery in carbonate reservoirs with complex permeability heterogeneity.

Introduction

The Shuaiba Formation, a prolific Lower Cretaceous carbonate reservoir in Abu Dhabi, is characterized by significant heterogeneity in depositional facies, diagenetic overprinting, and structural complexity (Fig.1.). Despite extensive development, reservoir performance has been impacted by early water breakthrough, poor sweep efficiency, and uneven waterflood conformance – largely attributed to the presence of high-permeability flow zones (HPFZ). HPFZs exhibit permeabilities several orders of magnitude greater than the surrounding matrix and can act as thief zones, bypassing oil and undermining recovery efficiency.

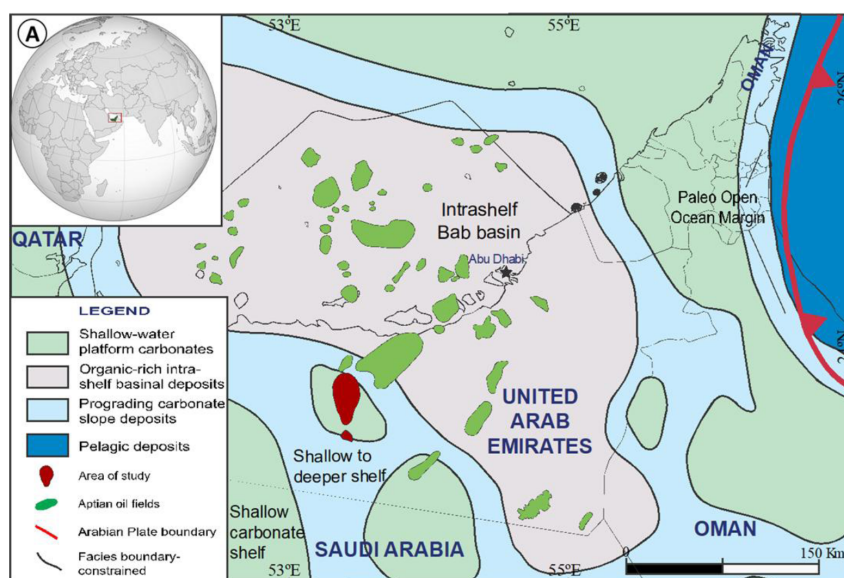


Figure 1—Paleogeographic map of the Shuaiba Formation during the Aptian age, illustrating the study area located on the western margin of the intra-shelf Bab Basin. The map encompasses Giant Oilfield A and Field B within Abu Dhabi, UAE (modified from Vahrenkamp, 2010).

Traditional static models often fail to capture the complex distribution of these high-flow conduits, especially when they are controlled by subtle stratigraphic, diagenetic, or structural features. Recent studies in Middle Eastern carbonate reservoirs highlight the critical role of laterally continuous, grain- or facies-dominated beds, fracture corridors, and vuggy porosity zones – collectively referred to as HPFZ- in disproportionately influencing production and injection behavior. However, their prediction and incorporation into reservoir models remains challenging due to their limited visibility in conventional well logs and seismic data.

This study addresses this gap by developing an integrated geological and dynamic workflow to identify, characterize, and predict HPFZ in the Shuaiba Formation of a giant carbonate oilfield (Cross, and Burchette, 2024). More than 1000 wells, including core data, borehole image logs, seismic interpretations, production logs, and well test data, were analyzed. High-resolution sedimentological analysis and sequence stratigraphy provided the framework to delineate depositional architecture and identify zones of lateral continuity favorable to HPFZ development. Structural interpretation from 3D seismic and borehole images enabled mapping of fault-related fracture corridors. Diagenetic features, such as leaching and touching-vug porosity, were incorporated using petrographic and dynamic evidence.

These datasets were integrated into a recent static and dynamic model, where HPFZ were deterministically distributed and validated against production response. The improved understanding of HPFZ geometry, origin, and impact of flow behavior led to enhanced well planning strategies that avoid high-flow thief intervals, improve sweep, and support optimized recovery.

This paper presents, a comprehensive predictive methodology for HPFZ identification and integration. The approach has direct implications for reservoir management in mature carbonate systems, particularly in regions near structural fault systems and early water breakthrough are most severe.

Geological Setting

The Shuaiba Formation is an Aptian-aged carbonate unit deposited across the Arabian Plate during a period of extensive marine transgression and platform development. In Abu Dhabi, it forms one of the most prolific hydrocarbon-bearing formations, comprising a complex mosaic of isolated and aggraded carbonate platforms, ramps, and intrashelf basins. The formation is genetically tied to the regional sequence stratigraphic framework bounded by maximum flooding surfaces K70 (Early Aptian) and K80 (Middle Aptian), defined by Sharland (2001, 2004, Fig.2.).

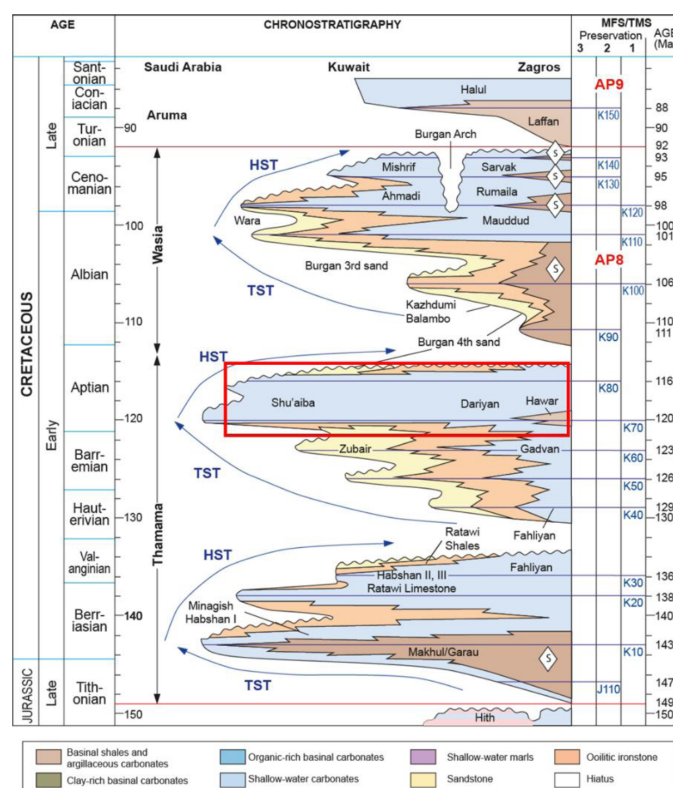


Figure 2—Chronostratigraphic section for megasequence AP8 (149-92 Ma). The Shuaiba Formation is within the second-order TST and HST of the middle sequence. Modified from Sharland et al. (2001, 2004)

The field geometry is a response to the Arabian Plate structural regime; it is defined as a low-relief asymmetrical anticline. Different hypotheses for its generation have been proposed, with the most likely being its relation to a deep-seated graben that underwent oblique inversion during the emplacement of the Oman ophiolites in the Late Cretaceous time.

The Shuaiba Formation records a progressive shallowing-upward trend and exhibits significant lateral and vertical facies heterogeneity. The studied field lies within a platform-ramp transition zone and is characterized by repeated regressive parasequences that form stacked, grain-rich rudist-bearing facies in the upper portions (Fig.3.). These facies often form laterally extensive, stratigraphic high-permeability flow zones (HPFZ) that exert a disproportionate control on fluid flow.

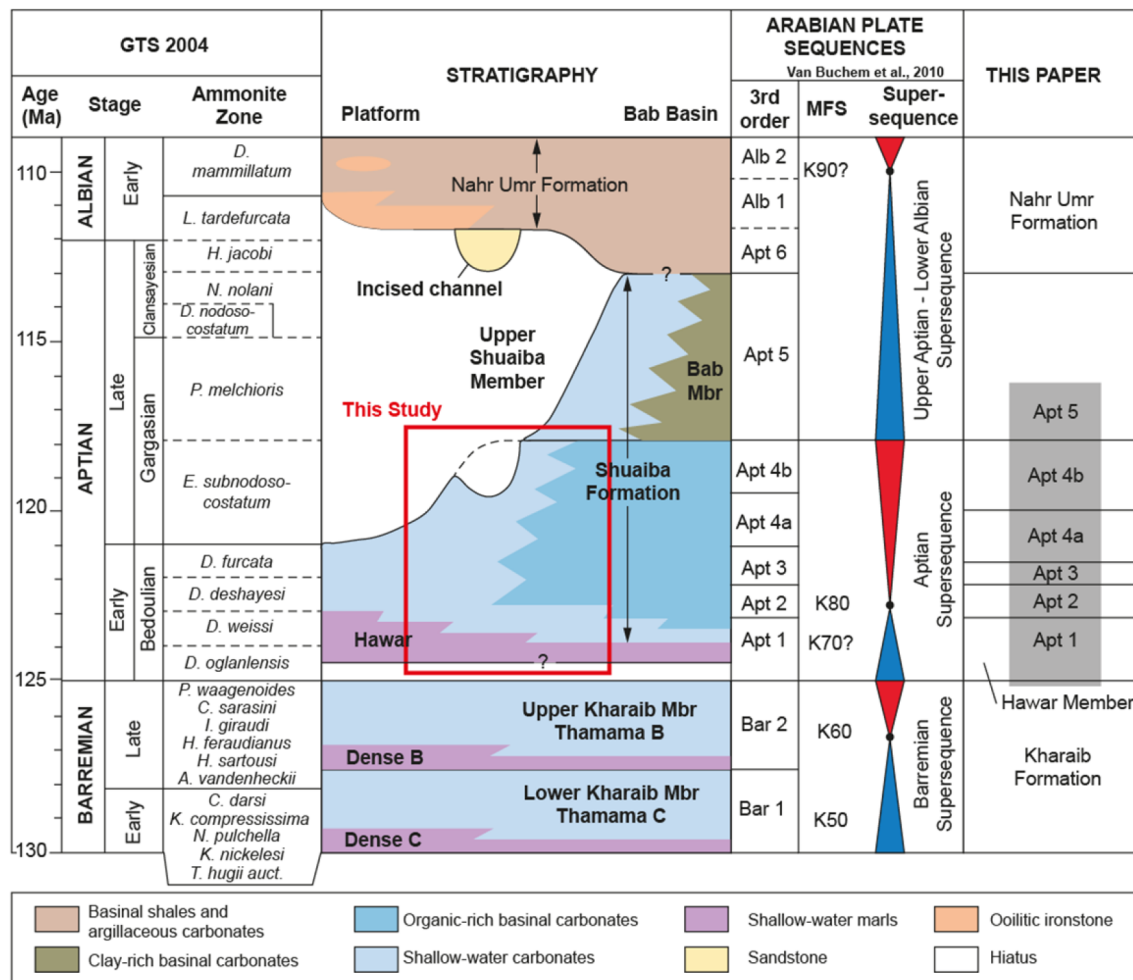


Figure 3—Schematic chart showing the five six-third-order depositional systems interpreted in the Aptian interval for the Arabian Plate (modified from van Buchem et al, 2010). The Shuaiba Formation comprises transition across the carbonate ramp with facies belts, inner to outer ramp to Bab Basin, and associated TST/HST system tracts as indicated by the red rectangle.

Facies distributions in the formation reflect a balance between carbonate productivity and accommodation space. The lower sections are dominated by mud-rich wackestones deposited under low-energy, restricted conditions. Upward transitions into peloidal packstones and rudist floatstones reflect increasing energy and shallower water depths. The grain-dominated, rudist-rich facies are the main contributors to reservoir quality and constitute the primary stratigraphic HPFZs. These layers are often aligned with regressive cycles and display continuity across several kilometers.

Diagenetic processes have further modified reservoir properties. Leaching and dissolution of aragonitic rudist shells, commonly associated with subaerial exposure or early meteoric diagenesis, have locally

enhanced porosity and permeability, creating diagenetic HPFZs. These diagenetic trends are often superimposed on the depositional architecture, increasing complexity in flow behavior.

Structurally, the field is dissected by low-offset faults, many of which remain subtle in seismic interpretation but are evident in borehole image logs and production behavior. These faults may form vertical corridors - structural HPFZs- that increase connectivity across flow units and can result in early water breakthrough, particularly in crestal wells. The combination of depositional architecture, diagenetic overprint, and structural features leads to a tripartite control over HPFZ development.

This geological framework provides the basis for integrating static and dynamic data to map and predict the distribution of HPFZs, a key step in optimizing waterflood strategies and improving recovery in this mature carbonate reservoir.

Methodology

A multidisciplinary workflow was applied to identify, characterize, and integrate High-Permeability Flow Zones (HPFZ) within the Shuaiba Formation using data from over 1,000 wells (Fig.4). This methodology combines sedimentological, stratigraphic, structural, and dynamic analyses to build a geologically consistent static model conditioned to fluid flow behavior. The following sub-sections describe each stage of the process (Fig.5).

- **Sedimentological Review of Core Data.** A core-based facies analysis was conducted on 150 key wells distributed across the field. Cores were re-described and classified according to dominant depositional textures and fauna, with particular attention to rudist-rich, grain-dominated facies. These facies, often associated with platform margin and high-energy shoal environments, were identified as the primary contributors to stratigraphic HPFZ development. Facies stacking patterns, indicative of regressive parasequences, were used to identify laterally continuous flow-contributing intervals. These interpretations provided the sedimentological foundation for mapping flow-contributing intervals across the field.
- **Refinement of Stratigraphic Surfaces.** A high-resolution stratigraphic correlation framework was built using well tops and core calibration. Surfaces such as sequence boundaries (SBs) and maximum flooding surfaces (MFSs) were refined and reinterpreted at multiple hierarchical scales to better delineate accommodation cycles and depositional trends (Magalhães et al., 2020). This allowed a more accurate positioning of facies associations and HPFZ within stratigraphic sequences. The framework also guided the reinterpretation of previously picked seismic horizons and facilitated vertical zonation for static modeling.
- **Seismic Horizon and Fault Interpretation.** Seismic horizons were revisited and refined in alignment with the updated stratigraphic surfaces. Fault interpretation was updated using both seismic coherence and curvature attributes. The new interpretation incorporated borehole image log data to validate fault orientations and displacement patterns. Structural elements were examined in relation to sequence stratigraphy to determine their influence on depositional patterns and to assess whether they enhanced or compartmentalized flow. Particular focus was placed on faults with associated fracture corridors contributing to structural HPFZ.
- **Borehole Image Log Standardization and Integration.** A total of 65 borehole image logs were standardized using a unified classification scheme for fracture type, orientation, and intensity. Image logs were tied back to the seismic fault framework to assess spatial relationships and validate the location of structural HPFZ. The lateral and vertical extent of influence of fault-related fracture corridors was estimated, supporting the mapping of potential vertical conduits. Fracture density maps were generated to calibrate structural inputs in the permeability model.
- **Integration of Production Logs (PLT).** Production logging data from several hundred wells were consolidated and interpreted to identify active producing intervals. These intervals were cross-

referenced with fracture-rich zones observed in borehole image logs. PLT inflow profiles were used to confirm the dynamic behavior of HPFZ, particularly those associated with structural features and high-energy facies. This integration confirmed that several zones with early water breakthrough and high fluid mobility coincide with mapped HPFZ, supporting their deterministic inclusion in the flow model.

- **Diagenetic Analysis and Core Calibration.** Petrographic data from thin section analysis were integrated with the stratigraphic surfaces defined in earlier steps. Diagenetic features—particularly leaching of rudist components and development of touching-vug porosity—were mapped and associated with high-frequency stratigraphic packages. These diagenetic HPFZs often overlay or cut across depositional layers, adding vertical heterogeneity to the permeability structure. Calibration with core and log-derived porosity-permeability relationships allowed incorporation of diagenetic overprints into the reservoir model.
- As part of the methodology, a bibliographic review was conducted to benchmark the approach against previous studies in carbonate reservoir characterization. Notable contributions include [Lapointe and Labourdette \(2012\)](#), who demonstrated the impact of stratigraphic architecture on permeability distribution; [Baghi et al. \(2020\)](#), who integrated seismic and petrophysical data for high-resolution modeling; [Li Wang et al. \(2021\)](#), who applied machine learning techniques to predict flow zones in heterogeneous carbonates; and [Carneiro et al. \(2015\)](#), who highlighted the importance of diagenetic overprints in reservoir performance. These studies provided valuable analogs and methodological insights that informed the workflow applied in this study.
- **Static Modeling and Permeability Architecture.** All observations were integrated into a unified static model. Discrete property models were developed to capture the distribution of HPFZ by assigning distinct facies and permeability multipliers based on their origin (stratigraphic, diagenetic, or structural). A permeability model was constructed that accounts for enhanced secondary permeability in HPFZ areas. This model served as input for dynamic simulations and was validated by matching field production trends and historical water breakthrough behavior.

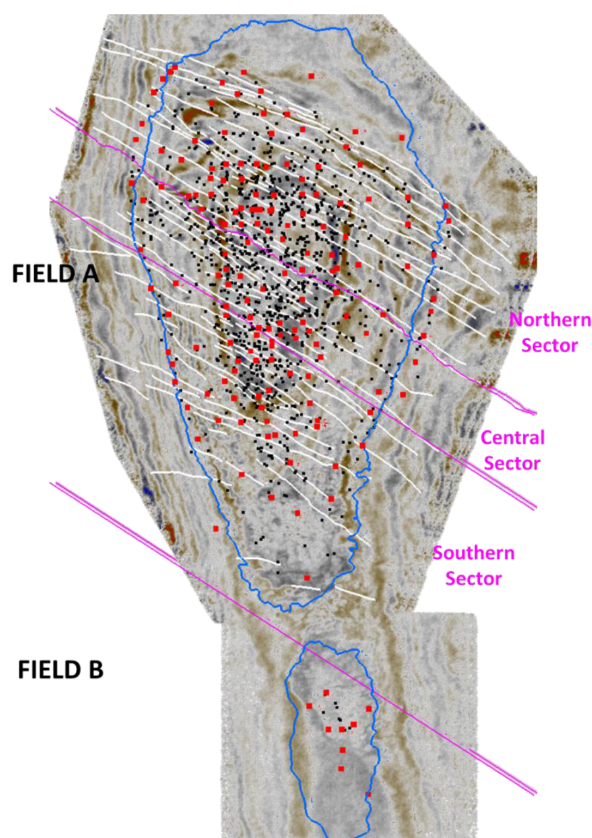


Figure 4—Dataset. Seismic coverage and structural framework of the study area. The continuous blue line delineates the boundaries of Oilfields A and B. Within Oilfield A, purple polygons segment the field into three regions: Southern, Central, and Northern sectors. White lines represent faults interpreted from seismic data. Black dots indicate the location of all drilled wells, while red dots highlight wells with available core data.

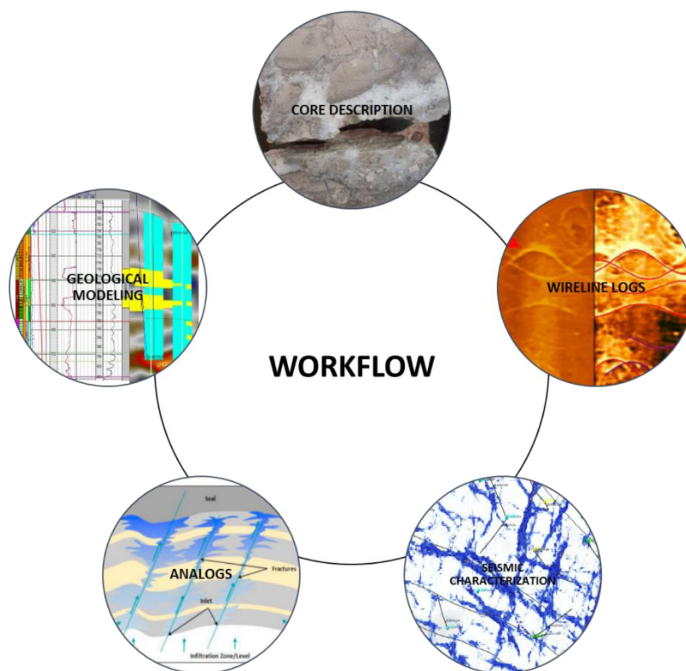


Figure 5—The methodology used in this study starts with the detailed description of structural features in cores (A), integration with wireline logs where conventional and image logs are included (B); seismic characterization (C), analogs review (D) and definition of HPFZs represented in 3D Geological Modeling process (E).

Results

The integrated workflow applied across the Shuaiba Formation yielded a comprehensive characterization of High-Permeability Flow Zones (HPFZs), revealing their diverse origins, spatial distribution, and impact on reservoir performance.

Stratigraphic HPFZs: Rudist-Rich and Regressive Facies

The revised high-resolution sequence stratigraphic (HRSS) framework subdivides the Aptian Shuaiba Formation into six third-order sequences: 1, 2, 3, 4a, 4b, and 5 (Fig.6). Each is bounded by well-defined sequence boundaries (SBs) and maximum flooding surfaces (MFSs) tied to regional stratigraphic markers. The updated correlations—based on >1,000 wells with core, wireline, seismic, and petrographic data—enable precise mapping of reservoir architecture and fluid flow barriers.

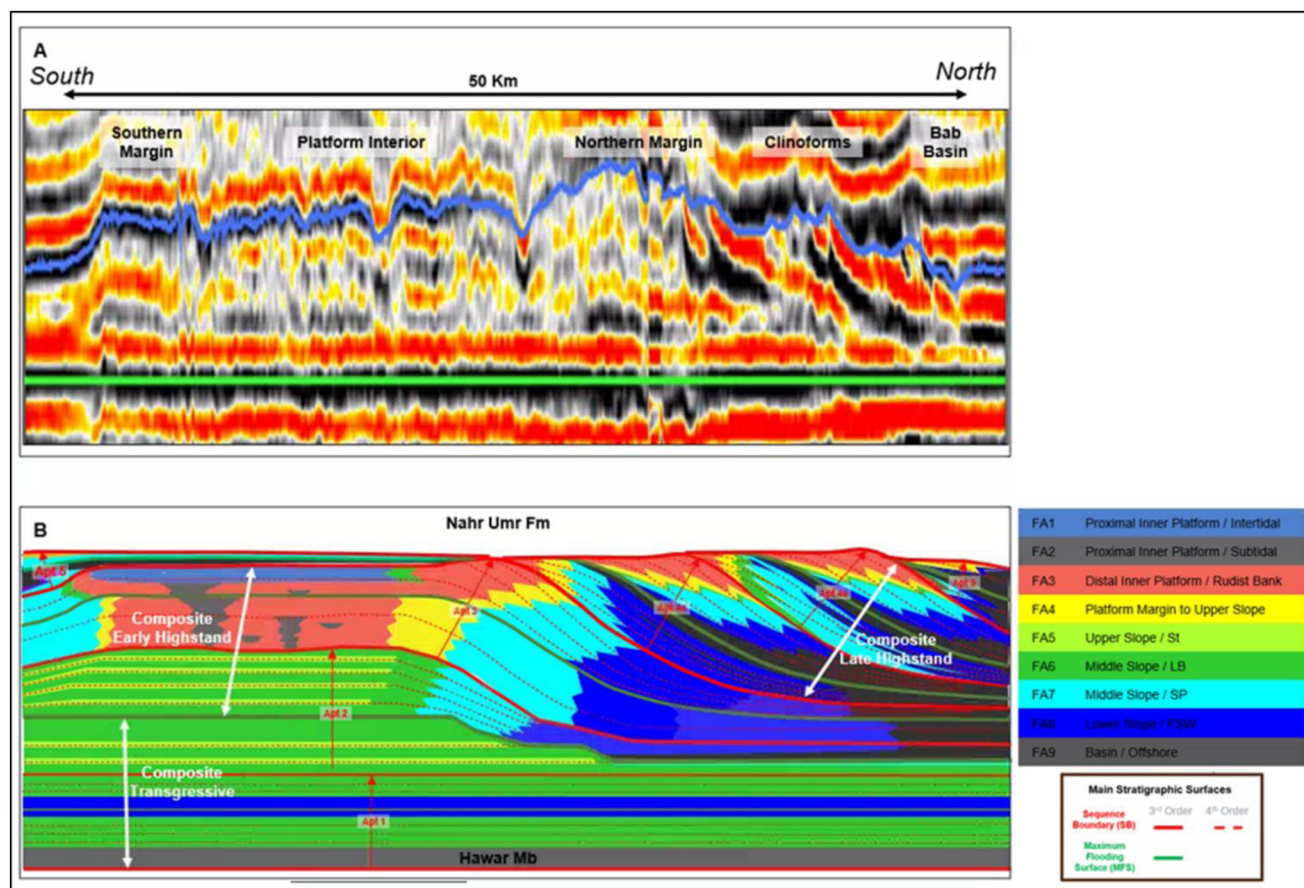


Figure 6—Modified Sequence-Stratigraphic model for onshore Abu Dhabi Field A and B. Above, S-N oriented seismic line showing the tabular geometries in southern margin, and platform area in contrast to clinoform geometries in northern margin, clinoform and Bab Basin areas. Below Facies Associations are divided into Second-order composites of Aptian Supersequence (White intervals) and Third-order sequences highlighted in red

Sequence Aptian 1 (AP1), this basal sequence, is characterized by mud-rich wackestone–packstone facies deposited under low-energy, restricted conditions. Reservoir quality is generally poor, except in localized rudist buildups near platform margins. The SB marks a shift from underlying marls to initial shallow-water carbonate accumulation. The overlying MFS is associated with finer facies, forming low-permeability intervals that act as baffles, particularly in the northern field.

Sequence Aptian 2 (AP2) represents a relative sea-level fall followed by rapid progradation of rudist shoal complexes. These grain-rich, high-energy deposits form one of the best HPFZ-bearing reservoir units

in the southern and central sectors. The MFS above this sequence is laterally persistent, capping the shoal bodies with muddy packstone to wackestone, which reduces vertical connectivity northward.

Sequence Aptian 3 (AP3) displays a mixed ramp system with alternating skeletal-peloidal packstone and mud-dominated facies. Reservoirs are moderate in quality, with HPFZs located in regressive shoal caps. The MFS coincides with a widespread flooding event depositing fine-grained, orbitolinid-bearing wackestones—laterally extensive in the north, forming a significant flow barrier between lower and upper reservoir packages.

Sequence Aptian 4a (AP4a) contains stacked rudist grainstone–packstone shoals aligned with regressive pulses. These host high-connectivity HPFZs that are laterally continuous across platform tops in the central field. The MFS of 4a is well-expressed in core as peloidal–skeletal wackestone with higher clay content, acting as a permeability barrier in the north.

Sequence Aptian 4b (AP4b) marks one of the most prolific reservoir intervals, with thick rudist-rich floatstone to grainstone bodies linked to platform margin progradation. Leaching and dissolution in rudist frameworks have created secondary porosity, enhancing HPFZ connectivity. The overlying MFS is mud-dominated, restricting vertical flow and controlling water movement during waterflooding.

Sequence Aptian 5 (AP5), represents the final phase of Shuaiba deposition before drowning. Reservoir quality diminishes upward as grain-rich facies give way to muddier ramp deposits. HPFZ development is localized to small, high-energy shoals, while the upper MFS forms a regionally extensive mud-rich cap that serves as the top seal for the formation.

Stratigraphic HPFZs were consistently associated with rudist-rich, grain-dominated facies deposited during regressive parasequences. Located near platform tops and ramp crests, these facies form laterally continuous layers with multi-Darcy permeability that contrast sharply with the surrounding micritic matrix (<10 mD). The correlation of these zones with high-frequency shallowing-upward cycles facilitated their deterministic mapping across the field. Core descriptions, poroperm data, and sequence stratigraphic surfaces confirmed that these HPFZs occupy predictable positions within the stratigraphic framework, often coinciding with stacked rudist shoals and cemented peloidal packstones.

Diagenetic HPFZs: Leached Zones with Enhanced Permeability

Microscale diagenetic analysis revealed the presence of leached intervals characterized by touching-vug porosity and dissolution-enhanced permeability (Fig. 7). These diagenetic HPFZs typically overprint primary depositional textures but show a strong stratigraphic control, often concentrated within exposure-prone or grain-rich facies reflecting high energy and shallower water depths. Thin section analysis, combined with core-calibrated porosity logs, enabled mapping of these features, especially in the southern area of the field, where diagenetic imprints were more pronounced. Their inclusion in the model significantly improved the vertical connectivity of flow units and matched local early breakthrough trends observed in production logs.

Core Description

Petrophysical Well Log Analysis

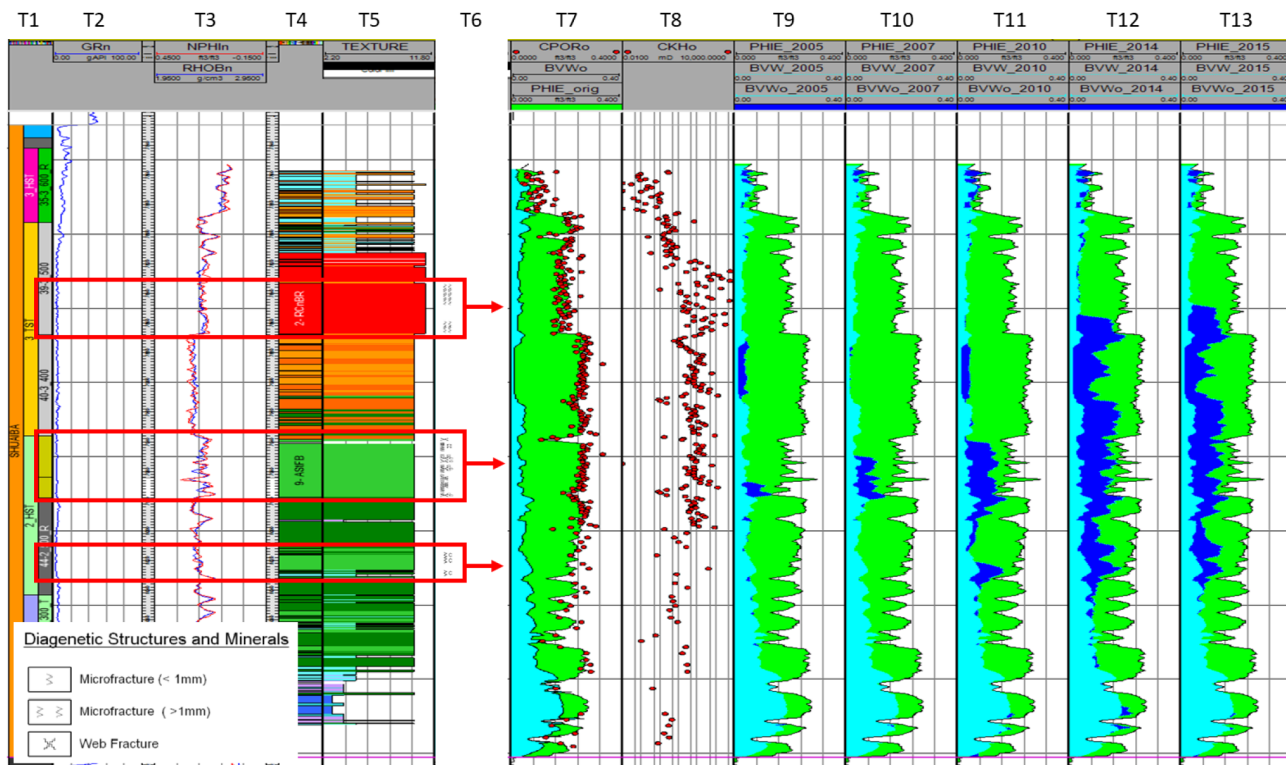


Figure 7—Integrated well section illustrating the correlation between core description and petrophysical analysis across 13 tracks (T1–T13). Tracks include: T1: Reservoir zone delineation; T2: Gamma ray log; T3: Density (blue) and neutron (red) logs; T4–T5: Lithofacies interpretation; T6: Description of natural fractures and fracture fill types; T7: Porosity log and core-derived porosity; T8: Permeability from conventional core analysis; T9–T13: Saturation profiles from RST logs, with water front highlighted in blue. Fracture-related water breakthrough intervals are marked with red rectangles, confirming the correlation between structural features and dynamic fluid response. Symbols indicate diagenetic features such as microfractures (<1 mm), macrofractures (>1 mm), and well-scale fractures, supporting the identification of HPFZs and their impact on reservoir performance.

Structural HPFZs: Fracture Corridors and Fault-Influenced Zones

The reinterpretation of seismic data and integration of 65 borehole image logs confirmed the presence of fracture corridors aligned with subtle fault systems (Fig.8). These fracture-dominated HPFZs exhibited vertical permeability enhancements, particularly in wells intersecting low-offset fault planes near the crest and graben-adjacent zones. Fracture density and orientation maps revealed preferential west-northwest trends consistent with interpreted fault strike. In several instances, PLT data confirmed high fluid mobility and early water influx in these corridors, validating their role as active thief zones (Fig.9).

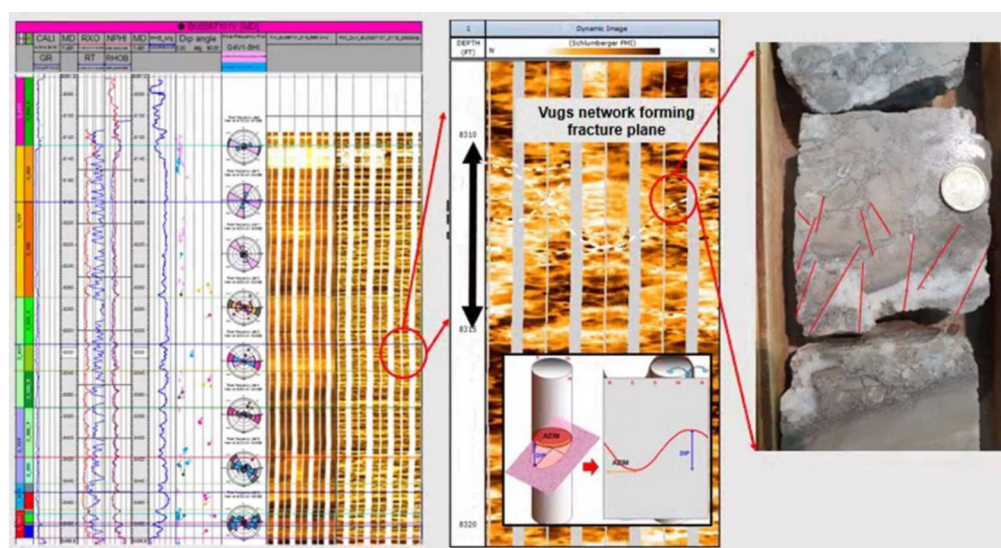


Figure 8—In cores description, identified early water breakthrough intervals are associated with vug porosity formed by partial dissolution of fracture networks, partially filled with calcite nodules. In image logs, interval appeared as conductive (dark) streaks and vugs (dark) surrounded by white resistive patches (blocky calcite).

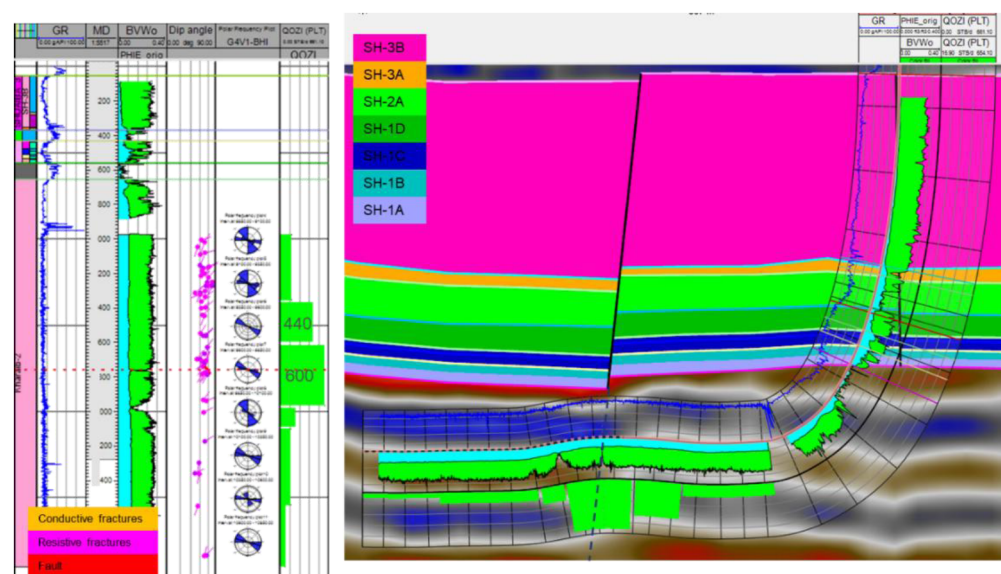


Figure 9—Integrated interpretation of seismic faults and borehole image logs used to delineate fracture damage zones proximal to fault planes. The left panel displays petrophysical and structural logs including Gamma Ray (GR), Porosity (Phi), dip angle, and polar frequency plot of fractures. Production Logging Tool (PLT) data are overlaid to highlight intervals of elevated fluid flow. The right panel presents a stratigraphic cross-section with units SH-3B to SH-1A, illustrating the spatial relationship between interpreted faults, fracture corridors, and dynamic flow behavior. This integration confirms that zones of high fluid mobility are frequently aligned with fault-associated fracture networks.

Several faults have been interpreted in the NW-SE trend that cuts across the axis of the structure. Like many carbonate reservoirs, it is possible to identify sub-seismic fractures on core samples, image logs, and seismic data. The density distribution of these features is mainly located in the center of the structure and the eastern flank, where the graben faults are located and the maximum absolute fault displacement was measured. As mentioned before, the presence of fractures has a high impact on fluid behavior and reservoir dynamics, providing additional permeability to the primary matrix permeability. These areas are defined in this paper as High Permeability Fracture Zones (HPFZ).

At the reservoir scale, we used a combination of seismic attributes. The most important piece of data was the seismic anisotropy volume, which captures the velocity variation that occurs when the seismic wave

passes through a set of oriented fractures, causing a change in the amplitude and velocity that varies with azimuth. This can also be referred to as azimuthal anisotropy. The recognizable anomalies in this volume were in the central part of the anticline and in the graben areas, both zones where the maximum deformation was expected.

Dynamic Behavior and Model Calibration

The incorporation of stratigraphic, diagenetic, and structural HPFZs into the static model enabled a significantly improved match between simulated and observed production behavior (Fig.10). In particular, water breakthrough timing, water cut evolution, and coning behavior were better replicated once HPFZs were explicitly modeled. This integration clarified previously unexplained discrepancies in waterflood performance and highlighted areas of bypassed oil due to preferential channeling through high-flow zones (Fig.11).

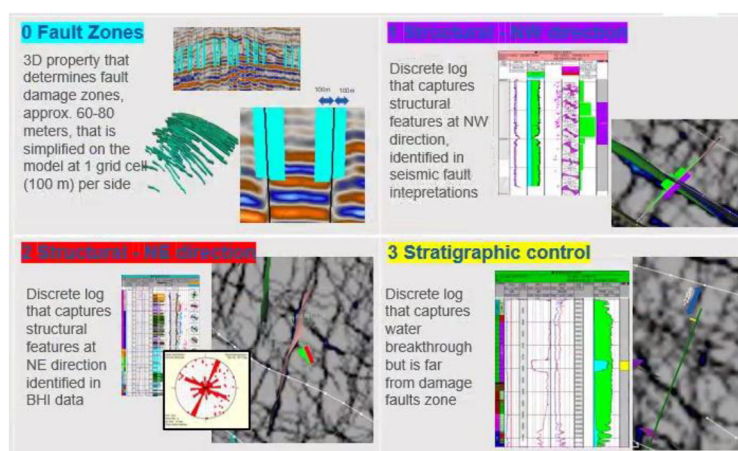


Figure 10—Legend that integrates the different classification HPFZs

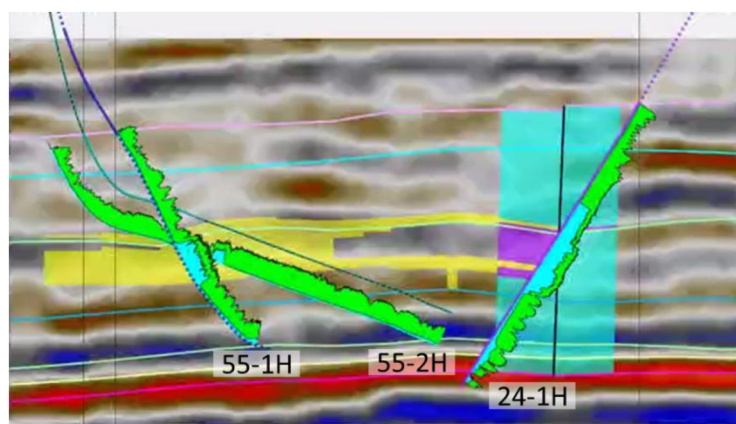


Figure 11—Cross-sectional view along a SW-NE orientation intersecting wells 55-1H, 55-2H, and 24-1H, which cross fault F34. Integration of borehole image logs enabled the identification and modeling of fracture damage zones, represented by light blue cells adjacent to seismic fault traces. Dynamic petrophysical log data were used to discretize intervals affected by water breakthrough, which were subsequently modeled as discrete properties (yellow cells) associated with stratigraphic controls. This integrated approach enhances the spatial understanding of fault-related permeability and its influence on fluid flow behavior.

Impact on Field Development Strategy

The updated model, conditioned by the integrated HPFZ framework, directly informed well placement and development decisions. Infill drilling strategies were optimized to avoid intersecting high-flow HPFZ corridors, particularly those associated with structural features and diagenetic vugs. Several proposed

injectors were repositioned to target zones with more favorable conformance behavior, while producers were redesigned to reduce early breakthrough risk. This new model supported the design of targeted completion strategies, including zonal isolation and selective stimulation, to mitigate the impact of HPFZs.

Discussion

The results of this study underscore the critical influence of High-Permeability Flow Zones (HPFZs) on the dynamic performance of the Shuaiba Formation. By integrating sedimentological, stratigraphic, structural, and diagenetic observations, it was possible to establish a predictive framework for their occurrence and quantify their impact on waterflood conformance.

Stratigraphic and Diagenetic Controls

The correlation between rudist-rich facies and regressive cycles confirms observations from other giant carbonate reservoirs in the Middle East, such as Ghawar and Zakum, where laterally continuous grain-dominated beds form preferential flow pathways. These stratigraphic HPFZs can extend over several kilometers and, when connected to wellbores, rapidly transmit injected water. The diagenetic overprint—particularly dissolution of rudist framework and matrix leaching—enhances permeability within these beds and creates vertical linkages that increase sweep bypassing. While the stratigraphic positioning of HPFZs makes them predictable, the superimposed diagenetic effects introduce local variability that requires core-calibrated mapping.

Structural Contribution to HPFZ Connectivity

Fault-related fracture corridors represent a separate but equally important category of HPFZ. In the studied field, fracture corridors coinciding with early water breakthrough zones indicate that even subtle structural features can provide significant vertical and lateral connectivity. The integration of borehole image logs with seismic fault mapping revealed that some fracture corridors extend beyond the seismic resolution, underscoring the importance of downhole imaging in mature fields. These corridors often link injectors and producers, especially in crest areas, accelerating water breakthrough and reducing sweep efficiency.

Implications for Reservoir Management

From a development standpoint, the explicit modeling of HPFZs improved the predictive capability of the dynamic simulation. Water cut trends, coning behavior, and breakthrough timing were better reproduced, allowing for more accurate forecasting of remaining oil distribution. This has direct implications for well planning:

- Producers can be placed in zones with lower HPFZ connectivity to maximize oil recovery before water arrival.
- Injectors can be reoriented or isolated from direct HPFZ contact to improve sweep patterns.
- Completion strategies can be adapted to selectively isolate HPFZ intervals using mechanical or chemical methods.

Broader Application and Limitations

The workflow demonstrated here is applicable to other mature carbonate fields with similar geological settings. Its success lies in the multi-scale integration of datasets—core, image logs, seismic, production logs, and diagenetic evidence—into a single, geologically consistent model. However, challenges remain in quantifying the permeability enhancement factor for fracture corridors and leached zones, as well as in capturing their potential changes over time due to compaction, scaling, or fines migration.

In conclusion, recognizing and explicitly modeling HPFZs—whether stratigraphic, diagenetic, or structural—is essential for accurate reservoir simulation and optimal field management. In the Shuaiba

Formation, this approach has directly influenced well placement, injection strategy, and long-term recovery optimization.

Conclusions

HPFZs as Key Flow Controls – High-Permeability Flow Zones (HPFZs) strongly influence fluid flow behavior and waterflood performance in the Shuaiba Formation, acting as dominant pathways for preferential water movement.

Stratigraphic Control – Rudist-rich, grain-dominated facies within regressive cycles form predictable, laterally continuous stratigraphic HPFZs, mainly in platform tops and ramp crest areas.

Diagenetic Enhancement – Meteoric-related leaching and dissolution of rudist framework generate diagenetic HPFZs with vertically enhanced permeability, increasing connectivity between flow units.

Structural Influence – Fault-related fracture corridors, often below seismic resolution, act as structural HPFZs, accelerating water breakthrough in crestal and fault-proximal wells.

Dynamic Model Improvement – Explicit inclusion of all HPFZ types in the static model significantly improved history matching, resolving discrepancies in waterflood performance.

Operational Impact – The refined model supports optimized well placement, selective completions to isolate HPFZ intervals, and improved injector positioning to enhance sweep efficiency.

Broader Applicability – This integrated workflow—combining sedimentological, stratigraphic, structural, and diagenetic analysis—offers a transferable methodology for HPFZ characterization in other mature carbonate reservoirs in the Middle East and globally.

Acknowledgments

The authors would like to thank the support to the oil company and their permission to publish this paper. This is part of a PhD project performed by the first author at the University of Barcelona (UB). The authors wish to thank all the colleagues that contributed to this paper.

References

- Peng, B., Zhang, H., Yang, S., Wang, S., Yang, Y., Shi, H., Hu, J., & Fang, X. (2020). Logging characterization of Carboniferous fractured-vuggy karst reservoirs in the eastern Qaidam Basin. *Journal of Geomechanics*, **26**(6), 923–931. <https://doi.org/10.12090/j.issn.1006-6616.2020.26.06.073>
- Carneiro, V. F. (2015). Caracterização da arquitetura e cinemática de falhas e fraturas e seu papel na distribuição da permeabilidade e da porosidade secundária em rochas carbonáticas carstificadas [Undergraduate thesis, Universidade Federal Fluminense]. Instituto de Geociências, UFF. Retrieved from http://www.geofisica.uff.br/sites/default/files/projetofinal/2012_vinicius_ferreira_carneiro.pdf
- Cross, N. E., & Burchette, T. (2024). Middle Eastern carbonate reservoirs – The critical impact of discrete zones of elevated permeability on reservoir performance. *Geological Society, London, Special Publications*, **548**. <https://doi.org/10.1144/sp548-2023-144>
- Lapointe, P. A., & Labourdette, R. (2012). Hydrothermal karst process-like modelling of Palaeozoic complex mound reservoirs from Kazakhstan. *AAPG Search and Discovery Article #90142*. Retrieved from <https://www.searchanddiscovery.com/abstracts/html/2012/90142ace/abstracts/lap.htm>
- Wang, L., Yang, R., Li, D., Meng, L., & Xiao, Z. (2021). Seismic attributes for characterization and prediction of carbonate faulted karst reservoirs in the Tarim Basin, China. *Interpretation*, **9**(3), T611–T622. <https://doi.org/10.1190/INT-2019-0162.1>
- Magalhães, A. J. C., Raja Gabaglia, G. P., Fragoso, D. G. C., Bento-Freire, E., Lykawka, R., Arregui, C. D., Silveira, M. M. L., Carpio, K. M. T., De Gasperi, A., Pedrinha, S., Artagão, V. M., Terra, G. J. S., Bunevich, R. B., Roemers-Oliveira, E., Gomes, J. P., Hernández, J. I., Hernández, R. M., & Bruhn, C. H. L. (2020). High-resolution sequence stratigraphy applied to reservoir zonation and characterisation, and its impact on production performance—Shallow marine, fluvial downstream, and lacustrine carbonate settings. *Earth-Science Reviews*, **210**, Article 103325. <https://doi.org/10.1016/j.earscirev.2020.103325>
- Sharland, P. R., Archer, R., Casey, D. M., Davies, R. B., Hall, S. H., Heward, A. P., Horbury, A. D., & Simmons, M. D. (2001). Arabian Plate sequence stratigraphy (GeoArabia Special Publication **2**, 371 p.). *Gulf PetroLink*.

- Sharland, P. R., Casey, D. M., Davies, R. B., Simmons, M. D., & Sutcliffe, O. E. (2004). Arabian Plate sequence stratigraphy—Revisions to SP2. *GeoArabia*, **9**(1), 199–214.
- van Buchem, F. S. P., Al-Shekaili, M. N., Droste, H. J., Maurer, F., Oterdoom, H. W., Vincent, B., Immenhauser, A., & Al-Mansouri, A. I. (2010). Barremian–Aptian stratigraphy and depositional systems of the Arabian Plate (GeoArabia Special Publication **4**). *Gulf PetroLink*